PG No. 15 D_{4h} $4/mmm$ [tetragonal] (internal axial octupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(3,3)}[g](A_{1u})$

** symmetry

1

** expression

$$\frac{\sqrt{105}G_{xyz}xyz}{7}+\frac{\sqrt{7}G_x^\alpha x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{105}G_x^\beta x\left(y-z\right)\left(y+z\right)}{14}-\frac{\sqrt{7}G_y^\alpha y\left(3x^2-2y^2+3z^2\right)}{14}$$
$$-\frac{\sqrt{105}G_y^\beta y\left(x-z\right)\left(x+z\right)}{14}-\frac{\sqrt{7}G_z^\alpha z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}G_z^\beta z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(3,1)}[g](A_{2g})$

** symmetry

z

** expression

$$\frac{\sqrt{21}G_{xyz}xy}{7}-\frac{3\sqrt{35}G_x^\alpha xz}{35}-\frac{\sqrt{21}G_x^\beta xz}{7}-\frac{3\sqrt{35}G_y^\alpha yz}{35}+\frac{\sqrt{21}G_y^\beta yz}{7}-\frac{3\sqrt{35}G_z^\alpha\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}G_z^\beta\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](A_{2g})$

** symmetry

z

** expression

$$-\frac{\sqrt{35}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{5\sqrt{21}G_x^\alpha xz\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta xz\left(2x^2+9y^2-5z^2\right)}{28}-\frac{5\sqrt{21}G_y^\alpha yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$-\frac{\sqrt{35}G_y^\beta yz\left(9x^2+2y^2-5z^2\right)}{28}+\frac{\sqrt{21}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}G_z^\beta\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

$\vec{\mathbb{G}}_{1,1}^{(3,1)}[g](E_g)$, $\vec{\mathbb{G}}_{1,2}^{(3,1)}[g](E_g)$

** symmetry

x

$-y$

** expression

$$\frac{\sqrt{21}G_{xyz}yz}{7}+\frac{3\sqrt{35}G_x^\alpha\left(2x^2-y^2-z^2\right)}{70}+\frac{\sqrt{21}G_x^\beta\left(y-z\right)\left(y+z\right)}{14}-\frac{3\sqrt{35}G_y^\alpha xy}{35}-\frac{\sqrt{21}G_y^\beta xy}{7}-\frac{3\sqrt{35}G_z^\alpha xz}{35}+\frac{\sqrt{21}G_z^\beta xz}{7}$$

$$-\frac{\sqrt{21}G_{xyz}xz}{7}+\frac{3\sqrt{35}G_x^\alpha xy}{35}-\frac{\sqrt{21}G_x^\beta xy}{7}+\frac{3\sqrt{35}G_y^\alpha\left(x^2-2y^2+z^2\right)}{70}+\frac{\sqrt{21}G_y^\beta\left(x-z\right)\left(x+z\right)}{14}+\frac{3\sqrt{35}G_z^\alpha yz}{35}+\frac{\sqrt{21}G_z^\beta yz}{7}$$

$\vec{\mathbb{G}}_{1,1}^{(3,3)}[g](E_g)$, $\vec{\mathbb{G}}_{1,2}^{(3,3)}[g](E_g)$

** symmetry

x

$-y$

** expression

$$\frac{\sqrt{35}G_{xyz}yz\left(6x^2-y^2-z^2\right)}{14}+\frac{\sqrt{21}G_x^\alpha\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{84}+\frac{\sqrt{35}G_x^\beta\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{28}$$
$$-\frac{5\sqrt{21}G_y^\alpha xy\left(3x^2-4y^2+3z^2\right)}{84}-\frac{\sqrt{35}G_y^\beta xy\left(5x^2-2y^2-9z^2\right)}{28}-\frac{5\sqrt{21}G_z^\alpha xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^\beta xz\left(5x^2-9y^2-2z^2\right)}{28}$$

$$\frac{\sqrt{35}G_{xyz}xz\left(x^2-6y^2+z^2\right)}{14}-\frac{5\sqrt{21}G_x^\alpha xy\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta xy\left(2x^2-5y^2+9z^2\right)}{28}-\frac{\sqrt{21}G_y^\alpha\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{84}$$
$$-\frac{\sqrt{35}G_y^\beta\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{28}+\frac{5\sqrt{21}G_z^\alpha yz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{\sqrt{35}G_z^\beta yz\left(9x^2-5y^2+2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_1^{(3,2)}[g](A_{2u})$

** symmetry

z

** expression

$$-\frac{\sqrt{105}G_{xyz}z\left(x-y\right)\left(x+y\right)}{14}+\frac{3\sqrt{7}G_{x}^{\alpha}y\left(4x^{2}-y^{2}-z^{2}\right)}{28}-\frac{\sqrt{105}G_{x}^{\beta}y\left(2x^{2}-y^{2}+z^{2}\right)}{28}$$
$$+\frac{3\sqrt{7}G_{y}^{\alpha}x\left(x^{2}-4y^{2}+z^{2}\right)}{28}+\frac{\sqrt{105}G_{y}^{\beta}x\left(x^{2}-2y^{2}-z^{2}\right)}{28}+\frac{\sqrt{105}G_{z}^{\beta}xyz}{7}$$

$\vec{\mathbb{Q}}_{1,1}^{(3,2)}[g](E_u)$, $\vec{\mathbb{Q}}_{1,2}^{(3,2)}[g](E_u)$
** symmetry

x

y

** expression

$$-\frac{\sqrt{105}G_{xyz}x\left(y-z\right)\left(y+z\right)}{14}+\frac{\sqrt{105}G_{x}^{\beta}xyz}{7}-\frac{3\sqrt{7}G_{y}^{\alpha}z\left(x^{2}-4y^{2}+z^{2}\right)}{28}$$
$$-\frac{\sqrt{105}G_{y}^{\beta}z\left(x^{2}+2y^{2}-z^{2}\right)}{28}+\frac{3\sqrt{7}G_{z}^{\alpha}y\left(x^{2}+y^{2}-4z^{2}\right)}{28}-\frac{\sqrt{105}G_{z}^{\beta}y\left(x^{2}-y^{2}+2z^{2}\right)}{28}$$

$$\frac{\sqrt{105}G_{xyz}y\left(x-z\right)\left(x+z\right)}{14}-\frac{3\sqrt{7}G_{x}^{\alpha}z\left(4x^{2}-y^{2}-z^{2}\right)}{28}-\frac{\sqrt{105}G_{x}^{\beta}z\left(2x^{2}+y^{2}-z^{2}\right)}{28}$$
$$+\frac{\sqrt{105}G_{y}^{\beta}xyz}{7}-\frac{3\sqrt{7}G_{z}^{\alpha}x\left(x^{2}+y^{2}-4z^{2}\right)}{28}+\frac{\sqrt{105}G_{z}^{\beta}x\left(x^{2}-y^{2}-2z^{2}\right)}{28}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_{2}^{(3,0)}[g](A_{1g})$
** symmetry

$$-\frac{x^{2}}{2}-\frac{y^{2}}{2}+z^{2}$$

** expression

$$-\frac{\sqrt{210}G_{xyz}\left(x-y\right)\left(x+y\right)}{28}-\frac{3\sqrt{14}G_{x}^{\alpha}yz}{28}-\frac{\sqrt{210}G_{x}^{\beta}yz}{28}+\frac{3\sqrt{14}G_{y}^{\alpha}xz}{28}-\frac{\sqrt{210}G_{y}^{\beta}xz}{28}+\frac{\sqrt{210}G_{z}^{\beta}xy}{14}$$

$\vec{\mathbb{Q}}_{2}^{(3,2)}[g](A_{1g})$
** symmetry

$$-\frac{x^{2}}{2}-\frac{y^{2}}{2}+z^{2}$$

** expression

$$\frac{\sqrt{105}G_{xyz}\left(x-y\right)\left(x+y\right)\left(x^{2}+y^{2}-6z^{2}\right)}{42}+\frac{5\sqrt{7}G_{x}^{\alpha}yz\left(6x^{2}-y^{2}-z^{2}\right)}{28}-\frac{\sqrt{105}G_{x}^{\beta}yz\left(12x^{2}-9y^{2}+5z^{2}\right)}{84}$$
$$+\frac{5\sqrt{7}G_{y}^{\alpha}xz\left(x^{2}-6y^{2}+z^{2}\right)}{28}+\frac{\sqrt{105}G_{y}^{\beta}xz\left(9x^{2}-12y^{2}-5z^{2}\right)}{84}-\frac{\sqrt{105}G_{z}^{\beta}xy\left(x^{2}+y^{2}-6z^{2}\right)}{21}$$

$\vec{\mathbb{Q}}_{2}^{(3,0)}[g](B_{1g})$
** symmetry

$$\frac{\sqrt{3}x^{2}}{2}-\frac{\sqrt{3}y^{2}}{2}$$

** expression

$$-\frac{\sqrt{70}G_{xyz}\left(x^{2}+y^{2}-2z^{2}\right)}{28}-\frac{\sqrt{42}G_{x}^{\alpha}yz}{28}+\frac{3\sqrt{70}G_{x}^{\beta}yz}{28}-\frac{\sqrt{42}G_{y}^{\alpha}xz}{28}-\frac{3\sqrt{70}G_{y}^{\beta}xz}{28}+\frac{\sqrt{42}G_{z}^{\alpha}xy}{14}$$

$\vec{\mathbb{Q}}_{2}^{(3,2)}[g](B_{1g})$
** symmetry

$$\frac{\sqrt{3}x^{2}}{2}-\frac{\sqrt{3}y^{2}}{2}$$

** expression

$$\frac{\sqrt{35}G_{xyz}\left(x^{4}-12x^{2}y^{2}+6x^{2}z^{2}+y^{4}+6y^{2}z^{2}-2z^{4}\right)}{42}+\frac{5\sqrt{21}G_{x}^{\alpha}yz\left(6x^{2}-y^{2}-z^{2}\right)}{84}+\frac{\sqrt{35}G_{x}^{\beta}yz\left(36x^{2}+y^{2}-13z^{2}\right)}{84}$$
$$-\frac{5\sqrt{21}G_{y}^{\alpha}xz\left(x^{2}-6y^{2}+z^{2}\right)}{84}-\frac{\sqrt{35}G_{y}^{\beta}xz\left(x^{2}+36y^{2}-13z^{2}\right)}{84}+\frac{5\sqrt{21}G_{z}^{\alpha}xy\left(x^{2}+y^{2}-6z^{2}\right)}{42}-\frac{\sqrt{35}G_{z}^{\beta}xy\left(x-y\right)\left(x+y\right)}{6}$$

$\vec{\mathbb{Q}}_{2}^{(3,0)}[g](B_{2g})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_x^\alpha xz}{7}+\frac{\sqrt{42}G_y^\alpha yz}{7}-\frac{\sqrt{42}G_z^\alpha (x-y)(x+y)}{28}+\frac{\sqrt{70}G_z^\beta \left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](B_{2g})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{35}G_{xyz}xy(x-y)(x+y)}{6}-\frac{5\sqrt{21}G_x^\alpha xz\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}G_x^\beta xz\left(2x^2-3y^2-z^2\right)}{12}-\frac{5\sqrt{21}G_y^\alpha yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^\beta yz\left(3x^2-2y^2+z^2\right)}{12}-\frac{5\sqrt{21}G_z^\alpha (x-y)(x+y)\left(x^2+y^2-6z^2\right)}{84}+\frac{\sqrt{35}G_z^\beta \left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{84}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,0)}[g](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(3,0)}[g](E_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$-\frac{\sqrt{42}G_x^\alpha (y-z)(y+z)}{28}-\frac{\sqrt{70}G_x^\beta \left(2x^2-y^2-z^2\right)}{28}-\frac{\sqrt{42}G_y^\alpha xy}{7}+\frac{\sqrt{42}G_z^\alpha xz}{7}$$

$$\frac{\sqrt{42}G_x^\alpha xy}{7}+\frac{\sqrt{42}G_y^\alpha (x-z)(x+z)}{28}+\frac{\sqrt{70}G_y^\beta \left(x^2-2y^2+z^2\right)}{28}-\frac{\sqrt{42}G_z^\alpha yz}{7}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,2)}[g](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(3,2)}[g](E_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{35}G_{xyz}yz(y-z)(y+z)}{6}+\frac{5\sqrt{21}G_x^\alpha (y-z)(y+z)\left(6x^2-y^2-z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta \left(4x^4-12x^2y^2-12x^2z^2+5y^4-18y^2z^2+5z^4\right)}{84}$$
$$+\frac{5\sqrt{21}G_y^\alpha xy\left(3x^2-4y^2+3z^2\right)}{84}+\frac{\sqrt{35}G_y^\beta xy\left(x^2-2y^2+3z^2\right)}{12}-\frac{5\sqrt{21}G_z^\alpha xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^\beta xz\left(x^2+3y^2-2z^2\right)}{12}$$

$$-\frac{\sqrt{35}G_{xyz}xz(x-z)(x+z)}{6}+\frac{5\sqrt{21}G_x^\alpha xy\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}G_x^\beta xy\left(2x^2-y^2-3z^2\right)}{12}+\frac{5\sqrt{21}G_y^\alpha (x-z)(x+z)\left(x^2-6y^2+z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^\beta \left(5x^4-12x^2y^2-18x^2z^2+4y^4-12y^2z^2+5z^4\right)}{84}+\frac{5\sqrt{21}G_z^\alpha yz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^\beta yz\left(3x^2+y^2-2z^2\right)}{12}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^\alpha x}{14}-\frac{\sqrt{35}G_x^\beta x}{14}-\frac{\sqrt{21}G_y^\alpha y}{14}+\frac{\sqrt{35}G_y^\beta y}{14}+\frac{\sqrt{21}G_z^\alpha z}{7}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^\alpha x\left(4x^2-21y^2+9z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta x\left(2x^2-3y^2-3z^2\right)}{28}+\frac{\sqrt{21}G_y^\alpha y\left(21x^2-4y^2-9z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^\beta y\left(3x^2-2y^2+3z^2\right)}{28}-\frac{\sqrt{21}G_z^\alpha z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A_{1u})$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{3\sqrt{770}G_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{462}G_x^{\alpha}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{1848}$$
$$-\frac{\sqrt{770}G_x^{\beta}x\left(2x^4+11x^2y^2-31x^2z^2+9y^4-87y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_y^{\alpha}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{1848}+\frac{\sqrt{770}G_y^{\beta}y\left(9x^4+11x^2y^2-87x^2z^2+2y^4-31y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}G_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{7}G_x^{\alpha}x}{14}-\frac{\sqrt{105}G_x^{\beta}x}{42}-\frac{3\sqrt{7}G_y^{\alpha}y}{14}-\frac{\sqrt{105}G_y^{\beta}y}{42}+\frac{\sqrt{105}G_z^{\beta}z}{21}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}G_x^{\alpha}x\left(4x^2-y^2-11z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{7}G_y^{\alpha}y\left(x^2-4y^2+11z^2\right)}{28}$$
$$-\frac{\sqrt{105}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{84}-\frac{5\sqrt{7}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{2310}G_{xyz}xyz\left(x-y\right)\left(x+y\right)}{44}+\frac{5\sqrt{154}G_x^{\alpha}x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_x^{\beta}x\left(2x^4-73x^2y^2+53x^2z^2+51y^4-87y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_y^{\alpha}y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_y^{\beta}y\left(51x^4-73x^2y^2-87x^2z^2+2y^4+53y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$
$$+\frac{\sqrt{2310}G_z^{\beta}z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z}{21}-\frac{\sqrt{7}G_x^{\alpha}y}{7}+\frac{\sqrt{105}G_x^{\beta}y}{21}-\frac{\sqrt{7}G_y^{\alpha}x}{7}-\frac{\sqrt{105}G_y^{\beta}x}{21}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{42}+\frac{\sqrt{7}G_x^{\alpha}y\left(4x^2-y^2-z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}y\left(6x^2+y^2-9z^2\right)}{84}$$
$$-\frac{\sqrt{7}G_y^{\alpha}x\left(x^2-4y^2+z^2\right)}{28}-\frac{\sqrt{105}G_y^{\beta}x\left(x^2+6y^2-9z^2\right)}{84}-\frac{5\sqrt{7}G_z^{\alpha}xyz}{7}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](B_{2u})$
** symmetry

$\sqrt{3}xy$

** expression

$$-\frac{\sqrt{2310}G_{xyz}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{462}+\frac{5\sqrt{154}G_x^{\alpha}y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{2310}G_x^{\beta}y\left(12x^4-46x^2y^2+66x^2z^2+5y^4-4y^2z^2-9z^4\right)}{924}+\frac{5\sqrt{154}G_y^{\alpha}x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{2310}G_y^{\beta}x\left(5x^4-46x^2y^2-4x^2z^2+12y^4+66y^2z^2-9z^4\right)}{924}-\frac{5\sqrt{154}G_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{3\sqrt{2310}G_z^{\beta}xyz\left(x-y\right)\left(x+y\right)}{44}$$

$\vec{\mathbb{G}}_{2,1}^{(3,-1)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(3,-1)}[g](E_u)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$\frac{\sqrt{105}G_{xyz}x}{21}-\frac{\sqrt{7}G_y^{\alpha}z}{7}+\frac{\sqrt{105}G_y^{\beta}z}{21}-\frac{\sqrt{7}G_z^{\alpha}y}{7}-\frac{\sqrt{105}G_z^{\beta}y}{21}$$

$$-\frac{\sqrt{105}G_{xyz}y}{21}+\frac{\sqrt{7}G_x^{\alpha}z}{7}+\frac{\sqrt{105}G_x^{\beta}z}{21}+\frac{\sqrt{7}G_z^{\alpha}x}{7}-\frac{\sqrt{105}G_z^{\beta}x}{21}$$

$\vec{\mathbb{G}}_{2,1}^{(3,1)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(3,1)}[g](E_u)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$-\frac{\sqrt{105}G_{xyz}x\left(2x^2-3y^2-3z^2\right)}{42}-\frac{5\sqrt{7}G_x^{\alpha}xyz}{7}-\frac{\sqrt{7}G_y^{\alpha}z\left(x^2-4y^2+z^2\right)}{28}$$
$$-\frac{\sqrt{105}G_y^{\beta}z\left(9x^2-6y^2-z^2\right)}{84}-\frac{\sqrt{7}G_z^{\alpha}y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}G_z^{\beta}y\left(9x^2-y^2-6z^2\right)}{84}$$

$$-\frac{\sqrt{105}G_{xyz}y\left(3x^2-2y^2+3z^2\right)}{42}-\frac{\sqrt{7}G_x^{\alpha}z\left(4x^2-y^2-z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}z\left(6x^2-9y^2+z^2\right)}{84}$$
$$+\frac{5\sqrt{7}G_y^{\alpha}xyz}{7}+\frac{\sqrt{7}G_z^{\alpha}x\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{105}G_z^{\beta}x\left(x^2-9y^2+6z^2\right)}{84}$$

$\vec{\mathbb{G}}_{2,1}^{(3,3)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(3,3)}[g](E_u)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$\frac{\sqrt{2310}G_{xyz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{462}+\frac{5\sqrt{154}G_x^{\alpha}xyz\left(2x^2-y^2-z^2\right)}{44}+\frac{3\sqrt{2310}G_x^{\beta}xyz\left(y-z\right)\left(y+z\right)}{44}$$
$$+\frac{5\sqrt{154}G_y^{\alpha}z\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}+\frac{\sqrt{2310}G_y^{\beta}z\left(9x^4-66x^2y^2+4x^2z^2-12y^4+46y^2z^2-5z^4\right)}{924}$$
$$+\frac{5\sqrt{154}G_z^{\alpha}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}-\frac{\sqrt{2310}G_z^{\beta}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{924}$$

$$\frac{\sqrt{2310}G_{xyz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{462}-\frac{5\sqrt{154}G_x^{\alpha}z\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{2310}G_x^{\beta}z\left(12x^4+66x^2y^2-46x^2z^2-9y^4-4y^2z^2+5z^4\right)}{924}+\frac{5\sqrt{154}G_y^{\alpha}xyz\left(x^2-2y^2+z^2\right)}{44}+\frac{3\sqrt{2310}G_y^{\beta}xyz\left(x-z\right)\left(x+z\right)}{44}$$
$$-\frac{5\sqrt{154}G_z^{\alpha}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}+\frac{\sqrt{2310}G_z^{\beta}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{924}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_3^{(3,-3)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$G_z^\alpha$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](A_{2g})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-G_{xyz}xy-\frac{\sqrt{15}G_x^\alpha xz}{20}-\frac{G_x^\beta xz}{4}-\frac{\sqrt{15}G_y^\alpha yz}{20}+\frac{G_y^\beta yz}{4}-\frac{\sqrt{15}G_z^\alpha\left(x^2+y^2-2z^2\right)}{15}-\frac{G_z^\beta(x-y)(x+y)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](A_{2g})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{330}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{132}-\frac{5\sqrt{22}G_x^\alpha xz\left(x^2-6y^2+z^2\right)}{44}+\frac{\sqrt{330}G_x^\beta xz\left(9x^2-12y^2-5z^2\right)}{132}+\frac{5\sqrt{22}G_y^\alpha yz\left(6x^2-y^2-z^2\right)}{44}$$
$$+\frac{\sqrt{330}G_y^\beta yz\left(12x^2-9y^2+5z^2\right)}{132}+\frac{\sqrt{22}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{88}+\frac{\sqrt{330}G_z^\beta(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](A_{2g})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{715}G_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{35\sqrt{429}G_x^\alpha xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_x^\beta xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144}$$
$$+\frac{35\sqrt{429}G_y^\alpha yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144}+\frac{7\sqrt{715}G_y^\beta yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}$$
$$-\frac{5\sqrt{429}G_z^\alpha\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_z^\beta(x-y)(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_{1g})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_{xyz}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_{1g})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-G_x^\alpha yz-G_y^\alpha xz-G_z^\alpha xy$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_{1g})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{7\sqrt{22}G_{xyz}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{66}-\frac{\sqrt{330}G_x^\alpha yz\left(6x^2-y^2-z^2\right)}{132}+\frac{35\sqrt{22}G_x^\beta yz(y-z)(y+z)}{132}$$
$$+\frac{\sqrt{330}G_y^\alpha xz\left(x^2-6y^2+z^2\right)}{132}-\frac{35\sqrt{22}G_y^\beta xz(x-z)(x+z)}{132}+\frac{\sqrt{330}G_z^\alpha xy\left(x^2+y^2-6z^2\right)}{132}+\frac{35\sqrt{22}G_z^\beta xy(x-y)(x+y)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_{1g})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{429}G_{xyz}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{286}$$
$$+\frac{7\sqrt{715}G_x^{\alpha}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}G_x^{\beta}yz\left(y-z\right)\left(y+z\right)\left(10x^2-y^2-z^2\right)}{572}$$
$$+\frac{7\sqrt{715}G_y^{\alpha}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}G_y^{\beta}xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572}$$
$$+\frac{7\sqrt{715}G_z^{\alpha}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{21\sqrt{429}G_z^{\beta}xy\left(x-y\right)\left(x+y\right)\left(x^2+y^2-10z^2\right)}{572}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_{2g})$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$G_z^{\beta}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_{2g})$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_x^{\alpha}xz}{4}-\frac{\sqrt{15}G_x^{\beta}xz}{4}-\frac{G_y^{\alpha}yz}{4}-\frac{\sqrt{15}G_y^{\beta}yz}{4}-\frac{G_z^{\alpha}\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_{2g})$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{35\sqrt{22}G_{xyz}xy\left(x-y\right)\left(x+y\right)}{132}+\frac{\sqrt{330}G_x^{\alpha}xz\left(5x^2+12y^2-9z^2\right)}{132}-\frac{5\sqrt{22}G_x^{\beta}xz\left(x^2-6y^2+z^2\right)}{132}-\frac{\sqrt{330}G_y^{\alpha}yz\left(12x^2+5y^2-9z^2\right)}{132}$$
$$+\frac{5\sqrt{22}G_y^{\beta}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{\sqrt{330}G_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{264}+\frac{7\sqrt{22}G_z^{\beta}\left(x^4-18x^2y^2+12x^2z^2+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_{2g})$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{429}G_{xyz}xy\left(x-y\right)\left(x+y\right)\left(x^2+y^2-10z^2\right)}{572}+\frac{7\sqrt{715}G_x^{\alpha}xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$
$$+\frac{7\sqrt{429}G_x^{\beta}xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144}-\frac{7\sqrt{715}G_y^{\alpha}yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$
$$-\frac{7\sqrt{429}G_y^{\beta}yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144}+\frac{7\sqrt{715}G_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$
$$-\frac{\sqrt{429}G_z^{\beta}\left(13x^6-45x^4y^2-150x^4z^2-45x^2y^4+540x^2y^2z^2+60x^2z^4+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{1144}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-3)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,-3)}[g](E_g,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

** expression

$$G_x^{\alpha}$$

$$-G_y^\alpha$$

$\vec{\mathbb{G}}_{3,1}^{(3,-3)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,-3)}[g](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$G_x^\beta$$

$$G_y^\beta$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

** expression

$$-G_{xyz}yz+\frac{\sqrt{15}G_x^\alpha\left(2x^2-y^2-z^2\right)}{15}-\frac{G_x^\beta(y-z)(y+z)}{2}-\frac{\sqrt{15}G_y^\alpha xy}{20}-\frac{G_y^\beta xy}{4}-\frac{\sqrt{15}G_z^\alpha xz}{20}+\frac{G_z^\beta xz}{4}$$

$$G_{xyz}xz+\frac{\sqrt{15}G_x^\alpha xy}{20}-\frac{G_x^\beta xy}{4}+\frac{\sqrt{15}G_y^\alpha\left(x^2-2y^2+z^2\right)}{15}-\frac{G_y^\beta(x-z)(x+z)}{2}+\frac{\sqrt{15}G_z^\alpha yz}{20}+\frac{G_z^\beta yz}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{G_x^\alpha(y-z)(y+z)}{2}+\frac{G_y^\alpha xy}{4}-\frac{\sqrt{15}G_y^\beta xy}{4}-\frac{G_z^\alpha xz}{4}-\frac{\sqrt{15}G_z^\beta xz}{4}$$

$$-\frac{G_x^\alpha xy}{4}-\frac{\sqrt{15}G_x^\beta xy}{4}+\frac{G_y^\alpha(x-z)(x+z)}{2}+\frac{G_z^\alpha yz}{4}-\frac{\sqrt{15}G_z^\beta yz}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{330}G_{xyz}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{\sqrt{22}G_x^\alpha\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{88}-\frac{\sqrt{330}G_x^\beta(y-z)(y+z)\left(6x^2-y^2-z^2\right)}{264}$$
$$-\frac{5\sqrt{22}G_y^\alpha xy\left(x^2+y^2-6z^2\right)}{44}-\frac{\sqrt{330}G_y^\beta xy\left(5x^2-9y^2+12z^2\right)}{132}-\frac{5\sqrt{22}G_z^\alpha xz\left(x^2-6y^2+z^2\right)}{44}+\frac{\sqrt{330}G_z^\beta xz\left(5x^2+12y^2-9z^2\right)}{132}$$

$$-\frac{\sqrt{330}G_{xyz}xz\left(x^2-6y^2+z^2\right)}{132}+\frac{5\sqrt{22}G_x^\alpha xy\left(x^2+y^2-6z^2\right)}{44}+\frac{\sqrt{330}G_x^\beta xy\left(9x^2-5y^2-12z^2\right)}{132}$$
$$-\frac{\sqrt{22}G_y^\alpha\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{88}+\frac{\sqrt{330}G_y^\beta(x-z)(x+z)\left(x^2-6y^2+z^2\right)}{264}$$
$$-\frac{5\sqrt{22}G_z^\alpha yz\left(6x^2-y^2-z^2\right)}{44}+\frac{\sqrt{330}G_z^\beta yz\left(12x^2+5y^2-9z^2\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E_g,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{35\sqrt{22}G_{xyz}yz\,(y-z)\,(y+z)}{132} - \frac{\sqrt{330}G_x^{\alpha}\,(y-z)\,(y+z)\left(6x^2-y^2-z^2\right)}{264} - \frac{7\sqrt{22}G_x^{\beta}\left(4x^4-12x^2y^2-12x^2z^2-y^4+18y^2z^2-z^4\right)}{264}$$
$$-\frac{\sqrt{330}G_y^{\alpha}xy\left(9x^2-5y^2-12z^2\right)}{132} - \frac{5\sqrt{22}G_y^{\beta}xy\left(x^2+y^2-6z^2\right)}{132} + \frac{\sqrt{330}G_z^{\alpha}xz\left(9x^2-12y^2-5z^2\right)}{132} - \frac{5\sqrt{22}G_z^{\beta}xz\left(x^2-6y^2+z^2\right)}{132}$$

$$-\frac{35\sqrt{22}G_{xyz}xz\,(x-z)\,(x+z)}{132} - \frac{\sqrt{330}G_x^{\alpha}xy\left(5x^2-9y^2+12z^2\right)}{132} - \frac{5\sqrt{22}G_x^{\beta}xy\left(x^2+y^2-6z^2\right)}{132} - \frac{\sqrt{330}G_y^{\alpha}\,(x-z)\,(x+z)\left(x^2-6y^2+z^2\right)}{264}$$
$$+\frac{7\sqrt{22}G_y^{\beta}\left(x^4+12x^2y^2-18x^2z^2-4y^4+12y^2z^2+z^4\right)}{264} + \frac{\sqrt{330}G_z^{\alpha}yz\left(12x^2-9y^2+5z^2\right)}{132} + \frac{5\sqrt{22}G_z^{\beta}yz\left(6x^2-y^2-z^2\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E_g,1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

** expression

$$\frac{7\sqrt{715}G_{xyz}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}$$
$$+\frac{5\sqrt{429}G_x^{\alpha}\left(16x^6-120x^4y^2-120x^4z^2+90x^2y^4+180x^2y^2z^2+90x^2z^4-5y^6-15y^4z^2-15y^2z^4-5z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_x^{\beta}\,(y-z)\,(y+z)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_y^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144} - \frac{7\sqrt{715}G_y^{\beta}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_z^{\alpha}xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144} + \frac{7\sqrt{715}G_z^{\beta}xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$

$$-\frac{7\sqrt{715}G_{xyz}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572} + \frac{35\sqrt{429}G_x^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_x^{\beta}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144}$$
$$+\frac{5\sqrt{429}G_y^{\alpha}\left(5x^6-90x^4y^2+15x^4z^2+120x^2y^4-180x^2y^2z^2+15x^2z^4-16y^6+120y^4z^2-90y^2z^4+5z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_y^{\beta}\,(x-z)\,(x+z)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_z^{\alpha}yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144} + \frac{7\sqrt{715}G_z^{\beta}yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E_g,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{21\sqrt{429}G_{xyz}yz\,(y-z)\,(y+z)\left(10x^2-y^2-z^2\right)}{572} + \frac{7\sqrt{715}G_x^{\alpha}\,(y-z)\,(y+z)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$+\frac{\sqrt{429}G_x^{\beta}\left(8x^6-60x^4y^2-60x^4z^2+150x^2y^4-540x^2y^2z^2+150x^2z^4-13y^6+45y^4z^2+45y^2z^4-13z^6\right)}{1144}$$
$$+\frac{7\sqrt{715}G_y^{\alpha}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144} + \frac{7\sqrt{429}G_y^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_z^{\alpha}xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144} + \frac{7\sqrt{429}G_z^{\beta}xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144}$$

$$\frac{21\sqrt{429}G_{xyz}xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572}-\frac{7\sqrt{715}G_x^{\alpha}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$+\frac{7\sqrt{429}G_x^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144}-\frac{7\sqrt{715}G_y^{\alpha}\left(x-z\right)\left(x+z\right)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$-\frac{\sqrt{429}G_y^{\beta}\left(13x^6-150x^4y^2-45x^4z^2+60x^2y^4+540x^2y^2z^2-45x^2z^4-8y^6+60y^4z^2-150y^2z^4+13z^6\right)}{1144}$$
$$+\frac{7\sqrt{715}G_z^{\alpha}yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}-\frac{7\sqrt{429}G_z^{\beta}yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}G_x^{\alpha}y}{4}-\frac{\sqrt{5}G_x^{\beta}y}{4}+\frac{\sqrt{3}G_y^{\alpha}x}{4}-\frac{\sqrt{5}G_y^{\beta}x}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{10}G_{xyz}z\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{6}G_x^{\alpha}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_x^{\beta}y\left(x-z\right)\left(x+z\right)}{4}$$
$$-\frac{\sqrt{6}G_y^{\alpha}x\left(2x^2-3y^2-3z^2\right)}{12}+\frac{\sqrt{10}G_y^{\beta}x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{10}G_z^{\beta}xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{55}G_{xyz}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{33}G_x^{\alpha}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}$$
$$+\frac{\sqrt{55}G_x^{\beta}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}-\frac{5\sqrt{33}G_y^{\alpha}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}$$
$$-\frac{\sqrt{55}G_y^{\beta}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}-\frac{7\sqrt{55}G_z^{\beta}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_x^{\beta}x}{3}-\frac{\sqrt{3}G_y^{\beta}y}{3}-\frac{\sqrt{3}G_z^{\beta}z}{3}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{10}G_x^{\alpha}x\left(y-z\right)\left(y+z\right)}{4}-\frac{\sqrt{6}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_y^{\alpha}y\left(x-z\right)\left(x+z\right)}{4}$$
$$+\frac{\sqrt{6}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{7\sqrt{55}G_x^\alpha x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44}+\frac{\sqrt{33}G_x^\beta x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}$$
$$+\frac{7\sqrt{55}G_y^\alpha y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44}+\frac{\sqrt{33}G_y^\beta y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}$$
$$-\frac{7\sqrt{55}G_z^\alpha z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{33}G_z^\beta z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_{xyz}z}{3}+\frac{\sqrt{5}G_x^\alpha y}{4}+\frac{\sqrt{3}G_x^\beta y}{12}+\frac{\sqrt{5}G_y^\alpha x}{4}-\frac{\sqrt{3}G_y^\beta x}{12}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_x^\alpha y\left(x-z\right)\left(x+z\right)}{4}+\frac{\sqrt{6}G_x^\beta y\left(3x^2-2y^2+3z^2\right)}{12}$$
$$+\frac{\sqrt{10}G_y^\alpha x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{6}G_y^\beta x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_z^\alpha xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{33}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}+\frac{\sqrt{55}G_x^\alpha y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$
$$-\frac{\sqrt{33}G_x^\beta y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}+\frac{\sqrt{55}G_y^\alpha x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$
$$+\frac{\sqrt{33}G_y^\beta x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}+\frac{7\sqrt{55}G_z^\alpha xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E_u,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{3}G_y^\alpha z}{4}-\frac{\sqrt{5}G_y^\beta z}{4}+\frac{\sqrt{3}G_z^\alpha y}{4}-\frac{\sqrt{5}G_z^\beta y}{4}$$

$$\frac{\sqrt{3}G_x^\alpha z}{4}-\frac{\sqrt{5}G_x^\beta z}{4}-\frac{\sqrt{3}G_z^\alpha x}{4}-\frac{\sqrt{5}G_z^\beta x}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E_u,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2}-\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{3}G_{xyz}x}{3}+\frac{\sqrt{5}G_y^\alpha z}{4}+\frac{\sqrt{3}G_y^\beta z}{12}+\frac{\sqrt{5}G_z^\alpha y}{4}-\frac{\sqrt{3}G_z^\beta y}{12}$$

$$-\frac{\sqrt{3}G_{xyz}y}{3}-\frac{\sqrt{5}G_{x}^{\alpha}z}{4}+\frac{\sqrt{3}G_{x}^{\beta}z}{12}-\frac{\sqrt{5}G_{z}^{\alpha}x}{4}-\frac{\sqrt{3}G_{z}^{\beta}x}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,0)}[g](E_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(3,0)}[g](E_u,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{10}G_{xyz}x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{10}G_{x}^{\beta}xyz}{2}-\frac{\sqrt{6}G_{y}^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{10}G_{y}^{\beta}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}G_{z}^{\alpha}y\left(3x^2-2y^2+3z^2\right)}{12}-\frac{\sqrt{10}G_{z}^{\beta}y\left(x-z\right)\left(x+z\right)}{4}$$

$$\frac{\sqrt{10}G_{xyz}y\left(x-z\right)\left(x+z\right)}{4}+\frac{\sqrt{6}G_{x}^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_{x}^{\beta}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_{y}^{\beta}xyz}{2}+\frac{\sqrt{6}G_{z}^{\alpha}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_{z}^{\beta}x\left(y-z\right)\left(y+z\right)}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,0)}[g](E_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(3,0)}[g](E_u,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2}-\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{6}G_{xyz}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_{x}^{\alpha}xyz}{2}-\frac{\sqrt{10}G_{y}^{\alpha}z\left(x-y\right)\left(x+y\right)}{4}$$
$$+\frac{\sqrt{6}G_{y}^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{10}G_{z}^{\alpha}y\left(x-z\right)\left(x+z\right)}{4}-\frac{\sqrt{6}G_{z}^{\beta}y\left(3x^2-2y^2+3z^2\right)}{12}$$

$$\frac{\sqrt{6}G_{xyz}y\left(3x^2-2y^2+3z^2\right)}{12}-\frac{\sqrt{10}G_{x}^{\alpha}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}G_{x}^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$+\frac{\sqrt{10}G_{y}^{\alpha}xyz}{2}+\frac{\sqrt{10}G_{z}^{\alpha}x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{6}G_{z}^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,2)}[g](E_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(3,2)}[g](E_u,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$-\frac{7\sqrt{55}G_{xyz}x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44}+\frac{7\sqrt{55}G_{x}^{\beta}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$-\frac{5\sqrt{33}G_{y}^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264}-\frac{\sqrt{55}G_{y}^{\beta}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88}$$
$$+\frac{5\sqrt{33}G_{z}^{\alpha}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}-\frac{\sqrt{55}G_{z}^{\beta}y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$

$$-\frac{7\sqrt{55}G_{xyz}y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44}+\frac{5\sqrt{33}G_{x}^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264}$$
$$+\frac{\sqrt{55}G_{x}^{\beta}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88}-\frac{7\sqrt{55}G_{y}^{\beta}xyz\left(x^2-2y^2+z^2\right)}{22}$$
$$+\frac{5\sqrt{33}G_{z}^{\alpha}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}-\frac{\sqrt{55}G_{z}^{\beta}x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,2)}[g](E_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(3,2)}[g](E_u,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2}-\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{33}G_{xyz}x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}-\frac{7\sqrt{55}G_x^{\alpha}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$-\frac{\sqrt{55}G_y^{\alpha}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88}-\frac{\sqrt{33}G_y^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264}$$
$$-\frac{\sqrt{55}G_z^{\alpha}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}+\frac{\sqrt{33}G_z^{\beta}y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}$$

$$\frac{\sqrt{33}G_{xyz}y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}-\frac{\sqrt{55}G_x^{\alpha}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88}$$
$$-\frac{\sqrt{33}G_x^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264}-\frac{7\sqrt{55}G_y^{\alpha}xyz\left(x^2-2y^2+z^2\right)}{22}$$
$$-\frac{\sqrt{55}G_z^{\alpha}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}+\frac{\sqrt{33}G_z^{\beta}x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}$$